Stage 3 Report: Second Approach, GAN-Based Super-Resolution

Image Super-Resolution with Deep Generative Models · Computer Vision 2026 · 16 Oct 2026 Team: Iuliia Kurnaeva, Alena Petrenko, Amaliya Kharisova, Dmitrii Kuznetsov

Repository: https://github.com/Ephy01/Image-Super-Resolution-with-Deep-Generative-Models

1. New approach

The second approach is **Real-ESRGAN x4plus**, the first generative model in the project. It is the same RRDBNet as our Real-ESRNet baseline (23 residual-in-residual dense blocks, 16.7 M parameters), but it was trained as the generator of a GAN. We use the official pretrained weights, loaded into our own definition of the network, and evaluate it on the same 256 low-resolution dev inputs: 64 source images × 4 JPEG settings.

2. Motivation

- **It isolates the effect of generative training.** Real-ESRNet and Real-ESRGAN have the same network, the same training data and the same synthetic degradations. Real-ESRGAN starts from the Real-ESRNet weights and is then trained with extra perceptual and adversarial losses. Any difference between the two is caused by the loss alone, which is the core of our research question.
- **Expected improvement.** An L1 loss averages over all plausible high-resolution images and gives smooth output. A GAN loss pushes the output towards realistic textures instead. We expect a clearly lower LPIPS (better perceptual quality) and, following the perception–distortion trade-off, a lower PSNR and SSIM than Real-ESRNet.
- **It is cheap.** The network is unchanged, so runtime and memory are the same as the baseline.

3. Implementation: differences from the first approach

	Real-ESRNet (baseline)	Real-ESRGAN (new)
Network	RRDBNet, 16.7 M parameters	the same
Training data and degradations	DIV2K, Flickr2K, OST; blur, noise, resize, JPEG (quality 30–95)	the same
Training loss	L1	L1 + perceptual (VGG-19) + GAN (U-Net discriminator with spectral norm)
Initialisation	from the ESRGAN weights	from the Real-ESRNet weights
Weights	`RealESRNet_x4plus.pth`	`RealESRGAN_x4plus.pth`

- **Code changes.** `download_data.py` also fetches the Real-ESRGAN weights. `upscale.py` loads either weight file into the same RRDBNet (`load_rrdbnet`), so both models share one inference path: input 128×128, output 512×512, batch size 1, float32, no tiling. `evaluate.py` adds the paired difference Real-ESRGAN − Real-ESRNet.
- **Training setup.** Unchanged: nothing is trained or fine-tuned, everything runs on the dev split, and the test split is kept for the end.

4. Initial results and comparison with the baseline (dev split, 64 images)

PSNR and SSIM on the Y channel with a 4-pixel border crop; LPIPS with AlexNet. Mean over images. Best value per content type in bold.

Content	Method	PSNR ↑, no JPEG	SSIM ↑	LPIPS ↓	PSNR ↑, JPEG 10	SSIM ↑	LPIPS ↓
Objects	Bicubic	25.62	0.689	0.488	22.77	0.543	0.767
Objects	Real-ESRNet	**26.31**	**0.737**	0.376	**22.97**	**0.582**	0.613
Objects	Real-ESRGAN	24.44	0.682	**0.255**	21.80	0.547	**0.451**
Faces	Bicubic	33.40	0.879	0.289	28.25	0.768	0.647
Faces	Real-ESRNet	**33.76**	**0.894**	0.238	**28.72**	**0.802**	0.456
Faces	Real-ESRGAN	31.05	0.859	**0.159**	27.65	0.776	**0.342**
Text	Bicubic	16.46	0.707	0.436	15.12	0.581	0.639
Text	Real-ESRNet	**19.31**	**0.838**	0.186	**15.67**	**0.690**	0.408
Text	Real-ESRGAN	17.83	0.818	**0.107**	14.38	0.680	**0.212**

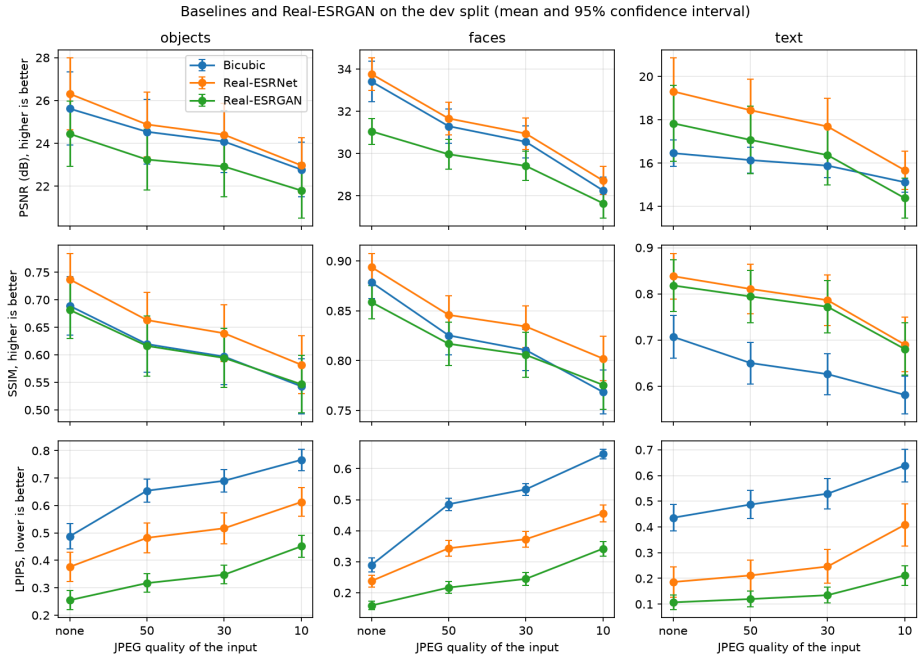
Paired difference Real-ESRGAN − Real-ESRNet on the same images, no JPEG, with 95% bootstrap interval: PSNR −1.87 ± 0.37 dB (objects), −2.71 ± 0.41 dB (faces), −1.47 ± 0.25 dB (text); LPIPS −0.12, −0.08, −0.08, that is 32%, 33% and 43% lower. Runtime on an NVIDIA RTX 3080 Ti Laptop GPU: bicubic 0.002 s, Real-ESRNet 0.11 s, Real-ESRGAN 0.11 s per image.

5. Problems and observations

- **The expected LPIPS–PSNR trade-off holds on 255 of the 256 image–JPEG pairs.** Real-ESRGAN has a lower LPIPS than Real-ESRNet on all 256 image × JPEG pairs and a lower PSNR on 255 of them. The 95% confidence intervals for the paired differences exclude zero for every content type and JPEG level.
- **On small text, LPIPS rewards invented letters.** With letters 4 px and 6 px tall in the input, Real-ESRGAN is 1.8 dB and 0.5 dB below bicubic in PSNR, yet it has the best LPIPS. Its letters are sharp but often different from the true ones (see the text row below). An OCR error rate is needed to measure this.
- **Faces lose the most fidelity.** The PSNR drop is largest on faces (−2.7 dB): the GAN adds skin texture and sharper eyes and teeth that are not in the original. Whether this changes identity will be measured with ArcFace.
- **Under heavy compression the GAN falls below bicubic.** At JPEG 10 the gap to Real-ESRNet shrinks to 1.1–1.3 dB, but Real-ESRGAN is now under bicubic in PSNR on all three content types. On text it turns compression blocks into confident, letter-like strokes instead of smoothing them.
- **Baseline numbers reproduced on another GPU.** We re-ran all methods on an NVIDIA GPU instead of the Apple M4. Per-image values differ from Stage 2 by at most 0.003 dB PSNR and 0.0004 LPIPS; after rounding, only one entry changes (faces, bicubic LPIPS 0.290 → 0.289).

6. Figures

Metrics against the JPEG quality of the input (mean and 95% confidence interval, dev split):



Central 256×256 crops, input without JPEG. The text row has letters 6 px tall in the input:

